



SOUTH
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COUNTRY CLIMATE AND DEVELOPMENT REPORT

Executive Summary

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Since independence, **Bangladesh has made significant progress on growth and poverty reduction**, with poverty declining from 48.9 percent in 2000 to 24.3 percent in 2016. Bangladesh has been among the fastest growing economies in the world, with annual per capita income growth of 4.0 percent over the past three decades. Further progress on human capital development, service delivery, connectivity, and rural welfare are imperative to sustain growth and poverty reduction. This must be accompanied by accelerated economic reforms to increase trade competitiveness, mobilize domestic revenue, deepen the financial sector, and strengthen climate resilience.

Bangladesh faces severe climate risks which, without strong action, will undermine its development trajectory with worsening impacts on the poor and vulnerable. Even under optimistic global climate scenarios, the country faces multiple severe risks from climate change. Average annual losses from tropical cyclones alone are already US\$1 billion per annum (0.7 percent of gross domestic product (GDP)). These impacts are likely to increase, including due to more frequent cyclone events that could result in significantly larger losses. Under severe flooding, GDP falls by as much as 9 percent relative to the baseline. Climate vulnerability is particularly acute on Bangladesh's coast. Under a 1.5 °C increase in temperature and 4 percent increase in precipitation, sea levels in the Bay of Bengal could rise by 27 centimeters or more by 2050. Such sea level rise would nearly double asset risk, currently at approximately US\$300 million per annum, and threaten agricultural production, water supply, and the diversity of coastal ecosystems. Without action, by 2040, cropland could shrink by more than 6.5 percent nationally and 18 percent in Southern Bangladesh. One-third of total agricultural GDP may be lost due to climate variability and extreme events by 2050. By this time, Bangladesh could see over 13 million internal climate migrants.

Climate change will disproportionately affect the poor, with impacts more concentrated in already vulnerable regions of Bangladesh. Building resilience with sustained and equitable growth is a development challenge. The costs of environmental degradation and natural disasters are expected to rise over time, compounded by higher heat and humidity and impacts on health. Investments to shore up resilience should be prioritized for hot spots such as in the Barind, Coastal zone, Haor area and Hill Tracts, and the three regions facing particularly high poverty rates and a compounding set of natural hazards: the western *upazilas* of Mymensingh, eastern *upazilas* of Rangpur, and southern parts of Khulna Division.

Furthermore, environmental degradation compounds the negative impacts of climate change. The annual cost of air pollution has been estimated at 9 percent of GDP while thirty-two percent of all deaths in Bangladesh are linked to environmental degradation, particularly outdoor and household air pollution; inadequate water, sanitation, and hygiene standards; and lead exposure. **Thus, the most far-reaching co benefits of climate action will come from improved air quality.**

These development challenges can be addressed while increasing resilience to climate risk. Transitioning away from high emission industries will positively impact poorer communities, including informal labor. Strengthening agricultural resilience to hotter, wetter conditions, safeguarding coastal areas and expanding nature-based solutions will support more equitable development by reducing rural stress and coastal erosion. And there is considerable opportunity for cleaner and more equitable growth through spatial development that leverages agglomeration effects and reduces congestion, makes cities more resilient to heat and reduces environmental pollution. Accelerating reforms and investments to address these risks will be foundational to sustaining a rapid pace of growth and poverty reduction.

Bangladesh recognizes these risks and opportunities and is committed to addressing them but needs to accelerate implementation. Managing climate risks and building resilience to cyclones, floods, and



other climate risks has been a hallmark of Bangladesh's economic development. Proactive policies and sound investments in strengthening resilience across multiple fronts over the last five decades have resulted in a drastic decline in the number of casualties from cyclones. Bangladesh has emerged as a global leader of the Vulnerable Twenty Group and has demonstrated global best practices on disaster risk management, such as cyclone preparedness. The country has also laid out a climate response vision in several plans and commitments, such as the Mujib Climate Prosperity Plan, Nationally Determined Contributions (NDC), National Adaptation Programme of Action, Bangladesh Delta Plan 2100 (BDP2100), 8th Five Year Plan, and Vision 2041.

This Country Climate and Development Report (CCDR) identifies near-term policy and investment priorities that will support Bangladesh to continue progress in building resilience to the effects of climate change. **Section 1** describes Bangladesh's vulnerability to the effects of climate change and outlines estimates of the cost of mitigation and adaptation investments through 2030. **Section 2** lays out the Government of Bangladesh's existing climate commitments and plans, and evaluates the institutional capacities required to meet them. **Section 3** highlights priority sector-level interventions to build climate resilience while meeting development goals. **Section 4** presents potential synergies between decarbonization and development. **Section 5** discusses the macroeconomic and distributional impacts of climate scenarios and identifies priority actions to support adaptation and growth. The CCDR provides additional analysis to prioritize actions to **accelerate** climate resilient development in line with Bangladesh's goals.

Priority Areas for Action

The CCDR has identified three interrelated priority areas for action.

Priority Area 1: People-centric, climate-smart spatial development

Accelerating the transition to a greener, more resilient, and more inclusive growth model will be crucial for development outcomes. Climate change will have disproportionate adverse impacts on the poor and vulnerable and moving towards more a climate-resilient development model would especially benefit these groups. Negative climate impacts could be substantially reduced through strategic adaptation investments. Key actions include increased efficiency and transformation of the agriculture and transport sectors, more climate-responsive social protection programs, investments in affordable, resilient housing, and expansion of nature-based solutions to protect coastlines and flood zones.

A climate-smart spatial approach to urban development is a priority. A spatially differentiated development strategy, built on urbanization as a driver of productivity and efficiency, can reduce regional disparities and urban poverty while increasing adaptation and resilience to climate change. This will be especially important given regional variations in climate impacts and the need for proactive planning for an increase in climate migrants to urban centers. Planning and budgeting for resilient transport connectivity, essential public services, water and waste management, affordable housing and housing retrofitting, resilient food systems, access to quality skills training and jobs, and appropriate social protection will help build resilience (including for internal migrants) to increasing disasters. Such investments need to be supported by institutional strengthening, including through the devolution of discretionary local government investment financing to *Union Parishad* levels and institutional alignment to drive efficient responses, such as the inclusion of the Ministry of Social Welfare in the National Disaster Management Council.

Strengthening the resilience of agriculture, water, and land use systems will contribute to further poverty reduction, especially in areas with high climate vulnerability and across urban and rural spaces. Agriculture has played an important role in Bangladesh's growth and poverty reduction, contributing to 69 percent of poverty reduction from 2005–10 and 27 percent from 2010–16. Including

livestock and fisheries, the sector represents around half of employment in the country and remains a significant source of livelihood for rural households. However, agriculture is severely threatened by the impacts of climate change on soil and water salinity, water stress, and temperature, as well as by unsustainable and inefficient use of inputs. Crop diversification and climate-smart technology adoption (with livestock and irrigation as priority sectors) and better linkages to market can together support higher agriculture productivity, better climate resilience, and higher rural incomes.

Implementing the Bangladesh Delta Plan 2100 (BDP2100) will promote sustainable management of water, ecology, environment, and land resources. It has been estimated that implementing the government's multisector BDP2100 as planned and in a timely manner could increase GDP by 25 percent over business as usual by 2041. The plan identifies "hotspots" for multisectoral investment, ranging from river training for flooding and transport to coastal resilience, agriculture transformation, water storage, and pollution treatment. The BDP2100 identifies investments of US\$38 billion by 2030 and has strong government commitment. The priority now is implementation. This requires prioritization of investment and policy needs; strengthening of the national legal framework; clarification and institutional arrangements to support implementation and cross-ministry coordination; strengthening, alignment, and diversification of financing, as well as an increase in development financing, domestic and foreign private sector investments. Together, these actions should translate the BDP2100's ambitious goals into the implementation of specific projects, committed and budgeted under Annual Development Plans.

Locally adapted and locally led solutions will also be essential to building resilience. Bangladesh has a rich history of locally led climate action and vibrant nongovernmental organizations that play a critical role in supporting communities and households. Integrating community-driven solutions into local development planning processes, building a robust understanding of local climate risks and uncertainties, and developing and ensuring consistent and predictable funding flows for locally led actions will continue to be critical.

The long-term low-carbon transition will have significant employment effects and social costs of adjustment, both of which will need to be actively managed. Low-carbon climate-resilient investments and structural changes in the economy are expected to have a positive effect on economy-wide labor demand, with job creation expected to occur broadly across economic sectors. Some decarbonization initiatives may adversely impact certain groups, however, and targeted transfers and special policy attention will be required to compensate the most vulnerable households.

Priority Area 2: Delivering development benefits with decarbonization

Emissions from energy, transport, industry, and agriculture can be reduced at relatively low-cost and yield significant cobenefits related to air pollution, health costs and jobs. Bangladesh currently contributes less than 0.4 percent to global greenhouse gas (GHG) emissions (221 million tons of carbon dioxide equivalent (MtCO₂e) in 2018) and, at 0.96 metric tons of CO₂ emissions per capita, it is still well below other countries. However, its large population (164.7 million) and projected fast economic growth rate with increasing incomes means that it could have a material impact on global emissions by the middle of the century if it does not take steps to manage emissions. Bangladesh's updated 2021 NDC commits to reducing emissions by 89.5 MtCO₂e, or 21.8 percent, relative to 2030 business as usual by 2030 of 409 MtCO₂e (including 6.7 percent in unconditional reductions and 15.1 percent conditional on international assistance). The analysis for this CCDR suggests that it is possible to meet these targets by 2030 without compromising development objectives. More importantly, investments to enhance energy efficiency, increase access to renewable energy, improve logistics to reduce congestion, and strengthen resilience of food and water systems are needed to support national development and growth objectives.



Decarbonization in the energy sector will deliver development benefits, limit lock-in to carbon-intensive investments, and mitigate the risk of stranded assets. The power sector has grown substantially in the last 15 years, increasing generating capacity from 4.5 gigawatts to 21 gigawatts and increasing access. However, power outages continue largely due to transmission bottlenecks, which costs the economy up to 1.5 percent of GDP per annum. The generation mix is overwhelmingly fossil-based, with natural gas at 70 percent and renewable energy less than 1 percent. The recent energy crisis in Bangladesh underlines the vulnerability of the sector due to such high dependency on fossil fuels. With a concerted effort, the power sector emission reductions outlined in Bangladesh's NDCs can be achieved.

Implementing the Energy Efficiency and Conservation Master Plan across all sectors is feasible and is a priority. Removing energy subsidies and domestic protection of inefficient industries will be critical to encouraging domestic and foreign investment in energy and resource efficiency. Enforcement of environmental standards must also be strengthened. Shifting the RMG, cement, steel, and fertilizer industries to a more sustainable path through energy and resource efficiency, integration of renewable energy, and compliance with environmental standards will support global competitiveness. Although heavy reliance on gas will continue with limited short- to medium-term alternatives, repurposing highly subsidized gas tariffs can reduce inefficient use of gas (and emissions), while the elimination of gas leaks in all parts of the system through technological and monitoring upgrades can significantly mitigate emissions. Reducing emissions beyond 2030 will require deeper action on energy efficiency, climate-smart grids, and greater domestic and imported renewable energy. Achieving these targets will depend on storage, carbon capture, and/or other alternative solutions that can be added over time to accommodate renewables in the power system. Bangladesh can monitor the evolution of clean and flexible energy solutions, including long-term storage and green hydrogen. Increased renewable energy imports from India, Nepal, and Bhutan offer significant mutual opportunities and should be accelerated. Power system development plans should be updated regularly as alternate power technologies develop and become financially competitive.

Transport is expected to account for up to 15 percent of emissions over the next decade, requiring a modal shift for emissions reduction. With economic growth, the continuing dominance of road transport will lead to higher congestion, emissions, and costs in the coming decades and will require a shift from road to rail and inland waterways as well as toward public and nonmotorized transport in urban areas. The introduction of electric vehicles offers a long-term opportunity—with the carbon benefit dependent on grid decarbonization. The National Logistics Development Policy, currently under development, presents opportunities to drive such transitions. Elimination of transport fuel subsidies and tax incentives that favor higher-carbon alternatives can help create incentives for change, including domestic and foreign private sector investment and innovation.

With rapid urbanization, income growth, and changing dietary patterns, opportunities exist for more efficient, low-carbon agriculture and food systems while increasing resilience and rural incomes. The agriculture sector, which contributes 37 percent of emissions, offers significant margins to reduce emissions. More efficient water management systems, market infrastructure, coordinated logistics, and integration of farmers in agro-supply chains—including female farmers who are a growing proportion of the population engaged in food production—are critical. Rapid urbanization and income growth have spurred a dietary shift toward nutrient-dense foods. This shift provides market opportunities for crop diversification so as to achieve high-value, climate-smart crops. Greater use of climate-smart agriculture practices and nature-based solutions can also reduce emissions while increasing resilience, reducing food loss and waste, improving livestock breeding and management practices, and providing alternative approaches to growing rice. Improved access to extension services and market intelligence to drive crop diversification and climate-smart technology adoption (with livestock and irrigation as priority sectors) can lead to emission reductions, as well as higher agriculture productivity and rural incomes.

Landscape restoration also can contribute to decarbonization and can support more productive systems through avoided deforestation, reforestation, and expansion of tree cover, thus delivering improved resilience for forest-dependent and coastal communities while expanding carbon sinks (a limited but not insignificant potential of up to 5–8 MtCO₂e per annum).

Priority Area 3: Enabling environment and institutional realignment

The most immediate challenge for Bangladesh is to improve legislative and institutional capacity for climate action and efficient implementation of existing policies and programs. The establishment of a Climate Change Act and high-level national coordination committee for national development and climate change will support alignment and transparent policy implementation. This high-level direction must strengthen the implementation strategy of the Bangladesh Climate Fiscal Framework 2020 and increase the regulatory capacity of the Ministry of Environment, Forests and Climate Change to support key ministries and departments such as energy, agriculture and water, to deliver on their climate programs. Empowering and capacitating local governments to plan and implement mitigation and adaption programs will improve efficiency and adaptation at the local level.

Overall, adaptation and decarbonization actions offer significant income and health cobenefits which are fully aligned with development priorities. Cross cutting institutional reforms can help realize these benefits. For example, demand- and supply-side energy efficiency can reduce overall costs, reduce waste, and increase income. Public investment in resilient infrastructure, NBS, public mass transit and modal shifts could alleviate potential pressure on existing public transport systems and reduce traffic congestion. Programs to improve livestock productivity and accelerate crop diversification can increase rural incomes.

Reducing tariffs and other barriers affecting trade in environmental goods and services can help Bangladesh participate in global value chains of such goods and services. Reforms to diversify exports in Bangladesh also must consider the challenges of graduating from least-developed-country status, increasing competition, changing demand patterns, and digitalizing opportunities. Lowering GHG emissions and strengthening environmental regulations will help, over time, to increase Bangladesh's competitiveness in global export markets, as well as improve the competitiveness of its domestic steel, cement, and fertilizer sectors.

There is considerable opportunity to expand the role of the private sector in climate action. While private sector participation in providing climate solutions remains at small scale, the BDP2100 and Mujib Climate Prosperity Plan expect 20–47 percent in financing from the private sector, amounting to approximately US\$1 billion and US\$7 billion, respectively. This is many times more than current private sector investment. Yet—as set out in this report—increased domestic and foreign private sector involvement, including in climate-smart agriculture, renewable energy and energy transmission, housing, and transport, is necessary and feasible. There is also considerable room for the private sector to lead on greening the ready-made garment sector (already under way), as well as in the heavy industrial cement, steel, and fertilizer sectors. With some effort, private participation in climate financing could rise to around 0.2 percent of GDP (up to US\$1 billion) by 2025.

Underlying financial sector vulnerabilities need to be addressed together with policies to green the financial sector. This includes measures to address growing financial sector vulnerabilities, including adequate recognition of non-performing loans (NPLs) and the associated capital adequacy of banks. New green financial products, internal capacity building, and improved regulatory incentives, together with addressing long-standing asset quality and governance weaknesses that distort efficient credit allocation, would set appropriate incentives to develop green finance and insurance markets. Long-term finance for climate adaptation and mitigation projects should be mobilized through capital market instruments (including sovereign and corporate green bonds) and participation of Bangladeshi firms in



carbon finance markets. At the same time, it is imperative to assess the climate-related physical and transitional risks faced by the financial sector and the economy at large, and to develop appropriate risk financing strategies and instruments including insurance and contingent financing mechanisms. Finally, from the demand side, the commercial viability of climate-smart projects can be improved by adopting less distortionary sector policies (e.g., cost recovery for energy, water and waste treatment, and other infrastructure services), along with the enforcement of improved regulatory frameworks, including the disclosure of climate impacts and risks.

Partnerships with NGOs to deliver and monitor climate change programs will be necessary, given their urgency, scale, and complexity. The Right to Information Act of 2009, parliamentary audits and accounts committees, an active court system, and a strong NGO advocacy provide the basis to establish a sound climate change accountability framework. Additional measures will facilitate this process, such as a climate change portal; parliamentary committee engagement on public accounts to analyze and report on climate-related public expenditures; and public data disclosure. A sustainable government procurement policy can drive a more inclusive and green public procurement framework with significant environmental and socioeconomic outcomes. Awareness, knowledge, and skills required for the transition to a low-carbon economy can be cultivated from basic education to reskilling. Demand-side accountability also can leverage the significant experiences of citizen engagement in Bangladesh for climate governance and a people-centric approach to adaptation. Youth and civil society-led accountability measures, together with a devolved climate finance structure for direct funding of youth groups and civil society organizations—will enable the creation of a diversified landscape of actors working to reduce vulnerability and enhance local area resilience.

Financing Needs and Availability

At a minimum, public and private financing options will need to support an estimated US\$12.5 billion in climate investment expenditure over the medium term (approximately 3 percent of FY21 GDP). Operationalizing financing at this scale will require strengthening public investment management and building an appropriate regulatory environment attract private investment. Reprioritizing the existing public investment envelope is the most accessible form of financing for the public sector, in the context of limited fiscal space. In FY21, capital expenditure stood at 5.1 percent of GDP. Prioritizing 20 percent of this budget for climate-related investments with significant co-benefits would provide approximately 1.0 percent of GDP. Over the longer term, strengthened domestic revenue mobilization and subsidy rationalization are needed to sustain a higher level of public investment, including administrative and policy reforms to limit tax expenditures and modernize consumption and income taxes.

Removal of subsidies, rationalization of gas tariffs, uniform value added tax on fuel transport via different modes, and a gradual introduction of a carbon tax will improve energy efficiency, reduce road congestion and generate revenue for redistributive and climate adaptation programs. Currently, user prices in many areas (e.g., fuel) are below optimal prices and, in some cases, do not even cover supply costs because of direct subsidies. The removal and/or repurposing of these subsidies could complement sectoral adjustments to improve productivity and climate responsiveness, while providing safety nets for the most vulnerable households. To further raise public sector finance (given the relatively low level of public domestic revenue mobilization), the GoB could consider (i) introducing a carbon tax and generate an additional US\$4.9 billion (1.2 percent of GDP, that could be split between investment and social protection); (ii) repurposing financing for subsidies to offset impacts and encourage climate investments (estimated at 1.6 percent of GDP); and (iii) redirecting investment financing to priority climate-resilience projects (1.0 percent of GDP). In energy, resources also can be mobilized from capital markets, based on the strength of SOE balance sheets, with the potential to generate an increase in commercial financing to US\$24 billion and US\$12 billion in transmission and distribution by 2041. The implementation of a carbon price with appropriate phasing and management

of the tough political economic issues associated with the rollout will not be easy, but a significant share of these resources, combined with a reprioritization of public investment, could be used to offset impacts on the poor and vulnerable while the balance of resources could be spent on accelerating adaptation or decarbonization as needed.

In addition, Bangladesh should continue its climate loss and damage advocacy and respond quickly to financing opportunities. This will require strong international partnerships and development partner financing, building on the needs and commitments of Bangladesh. At the same time, fiscal prudence and alternative planning should continue in case this funding is slow to materialize.

Prioritization

Given financial and institutional constraints, the most impactful and urgent actions to achieve climate resilient development must be prioritized. To do this, the CCDR presents a possible framework and an initial prioritization in the matrix below. This will require further refinement based on deeper consultation and discussion among key stakeholders in Bangladesh. The actions identified are those with the highest impact on development, adaptation, and mitigation with due consideration to trade-offs and distributional effects. They are prioritized based on urgency (adaptation to short-term risks and avoiding lock-in effects); and feasibility (readiness, existing commitments and political economy considerations, access to finance, and ease of implementation). Actions blue are the most urgent to adapt to current and short-term risks, avoid lock-in effects into carbon-intensive patterns, and minimize the accumulation of additional risks. Those in darker blue are at a higher level of readiness and those in lighter blue require additional effort to bring them closer to implementation. Actions in purple/pink are high impact but the risk of delayed implementation is lower. The ones in purple are at a high level of readiness while pink indicates a lower level of readiness.

Actions supporting adaptation and resilience are the highest priorities, given Bangladesh's high climate vulnerability, especially for the poor. On the mitigation side, in the near-term Bangladesh should prioritize policies and investments that deliver the highest development co-benefits (e.g. increased energy efficiency, better transmission, reduced gas losses, lowering air pollution and congestion and enabling renewable energy and imports to meet future energy needs). Focusing on these actions would enable Bangladesh to meet its NDC commitments through 2030. Options for deeper decarbonization can continue to be analyzed, and decisions can be taken when there is greater clarity on technology development and the availability of concessional international finance.



Bangladesh Country Climate and Development Report: Prioritization Matrix

	Urgency	
	High	Medium
Feasibility	Needs immediate action to address major vulnerabilities and prevent additional risks and lock-in effects.	Actions with lower lock-in risks.
High Identified in existing plans, with good level of technical and financial readiness.	Actions that should not be postponed and will produce a high impact. Delayed action will increase risk.	Actions with high level of readiness and impact without significant lock-in effects.
Medium Require additional resources/dialog to be ready for implementation.	Urgent actions with risk of delayed implementation that require additional dialog and resources	Actions with lower lock-in effects that need additional resources to be implemented.

 = Policies, Strategies, Institutions  = Investments

Policy Package 1: People-centric, Climate-smart Spatial Development

	Prioritize and allocate financing for Bangladesh Delta Plan 2100 projects. Projects identified in 2018 were estimated at US\$38 billion. Streamlining of institutional arrangements and improving public investment management are critical to accelerate implementation.
	Leverage financing mechanisms for locally led climate action. Increase awareness for local-level climate actions and strengthen systems for adaptation in a transparent and accountable manner. Current capacity levels of local government and communities to address climate vulnerabilities and work together may require further strengthening. Expand access to skills training and jobs for the urban poor and climate migrants.
	Realign education and skills training programs to raise climate awareness and meet skill needs for low-carbon transition. Technical training system to coordinate with industries and local governments to design and implement reskill training programs.
	Invest in the transformation of the agri-food sector. Requires repurposing agriculture support programs (fertilizer, water, energy subsidies); increasing investments in high-productivity, diversified, climate-smart livestock/crop systems; removing barriers for private sector (e.g., in seed production); and strengthening research-extension private sector linkages.
	Integrate climate migration into planning and development for select secondary cities and towns. Proactively plan for climate migration in selected areas. Need to address urban heat island risks and issues of vulnerability and marginalization.
	Scale up Nature-Based Solutions for resilience, carbon sequestration, ecosystem services, and livelihoods. Forest cover is presently carbon positive (0.37 MtCO ₂ e per annum). Restoration and afforestation will contribute to a reverse trend and make forests into carbon sink. Develop a joint regional investment plan (Bangladesh and India) for Resilient Sundarbans.
	Expand and improve cash-based shock-responsive social protection programs. Establish predetermined financing to increase budget predictability of shock response. Enhance efficiency by streamlining beneficiary identification, monitoring, and digital transfers. Leverage targeting and government-to-citizen (G2P) platforms to reach urban poor and climate migrants. Include the Ministry of Social Welfare in the National Disaster Management Council to increase efficiency for disaster response.
	Invest in energy-efficient and climate-resilient housing and urban infrastructure. Provide technical and financial support for new and existing affordable housing, prioritizing low-income communities and informal settlements. Implement urgent investments in water storage and incentives for conservation, and in solid waste management particularly in Greater Dhaka and Greater Chattogram. Leverage public-private partnerships.

Policy Package 2: Delivering Development Benefits with Decarbonization

	Implement energy efficiency and circular economy solutions in ready-made garments and textile factories and energy efficiency building standards. Target reductions of 30 percent with circular economy and energy efficiency measures. Requires regulation, coordination, and financing for private sector action. Will require short-term costs for long-term efficiency gains and competitiveness.
	Agriculture and Livestock Climate Smart investments and policies. To a) improve emission intensity in livestock systems; b) improve rice production and irrigation practices to reduce water and energy consumption and lower emissions and c) reduce food loss and waste across the food system through cooling solutions, improved logistics, storage and processing. Will require private sector and farmer aggregation solutions and adoption of standards for implementing cleaner production practices inside agro-processing facilities.

	Gradually eliminate implicit and explicit fossil fuel subsidies, including for gas. Requires related tariff reforms. Affordability and distributional impact need to be analyzed and targeted support for poor households may be required. Generates air quality co-benefits, incentivizes private-sector emission reductions and helps mitigate emissions in the gas value chain.
	Strengthen the power grid to optimize use of generation assets and integrate renewable energy (RE). Increase efficiency, improve flexibility, improve service, and integrate RE generation. Increase RE generation domestically and RE imports via regional power trade. Requires RE potential assessment, improved land-use planning, and government land allocation. Power trade requires regional cooperation, regulatory and contractual frameworks, and careful consideration of energy security.
	Introduce an energy focused carbon tax. A carbon tax would provide price incentives for mitigation while mobilizing modest additional revenues for climate investments. However, market-based energy pricing and further distributional analysis are required prior to implementation.
	Adopt National Logistics Development Policy and decrease road transport with increased transport by rail and waterways. Requires long-term infrastructure investment and consideration of the environmental impacts of river management. Electric vehicles offer a viable opportunity to reduce carbon emissions. Policies to support electric vehicle adoption may include tax rebates, flexible financing and related infrastructure, such as charging stations. Finalization and implementation of the proposed National Logistics Development policy with a focus on modal shift, resilient and low-carbon multimodal transport logistics and guidelines for private capital mobilization. This includes standardizing VAT rates on fuel transport via roads vs other transport modes.
Policy Package 3: Enabling Environment and Institutional Realignment	
	Develop and implement clean air programs to reach World Health Organization PM2.5 interim targets. Benefits to climate and air pollution alike through carbon dioxide, black carbon, and methane emission reductions. Support green tech and green financing uptake and help meet export standards.
	Establish a sustainable government procurement policy and implementation roadmap. Integrate procurement into the legal, regulatory, and institutional public procurement framework to incorporate environmental and socioeconomic considerations and enabling more inclusive and green public procurement.
	Establish a climate change portal with publicly disclosed data and engage parliamentary committee on public accounts to assess climate-related public expenditures and reports. Strong civil society and nongovernment organizational presence, as well as advocacy platforms to provide basis to support accountability on climate action.
	Strengthen assessment and disclosure of climate and environmental risks in financial and real sector. Physical and transitional climate risks in the banking sector require tools for assessment and mitigation, with prudential incentives for green lending. Disclosure requirements can help channel financing for climate investments.
	Reduce tariffs and behind-the-border barriers on environmental goods, based on clear classification of such goods. Ensure transparency by publishing the classification and providing advance rulings. Requires integration in a strategy of broad tariff modernization with low, uniform rates and elimination of para-tariffs.
	Enact a climate change act to align policies and programs between line ministries and establish a high-level national coordination committee. Provides high-level mandate for climate change prioritization and a mechanism for policy implementation, including leadership of the climate fiscal framework.
	Develop a risk financing strategy and instruments to mobilize additional resources and cover contingent liabilities. Adopt innovative strategies for climate finance, including land-based financing and risk insurance. Ministry of Finance to identify government priorities and implement a strategy with a comprehensive risk-layering approach to cover the government's explicit and implicit liabilities. Explore risk financing instruments to crowd in private sector. Green bonds/green <i>sukuk</i> can help raise long-term funding. Green bond guidelines needed for sovereign and private issuances, consistent with international standards. Leverage spatial risk assessment data and prioritize programs that target the poor, including property insurance.



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